Chapter 3

Cosmic-ray transport

CRs bombard the Earth steadily from all directions in the sky. We can detect these particles with space-based telescopes or detect the secondary particles they produce when entering the atmosphere with ground-based telescopes. These measurements provide us with the intensity and spectral energy distribution of CRs in the vicinity of the Earth (e.g. data from *PAMELA* and *AMS-02*) or at the furthest in the very local interstellar medium (data from the *Voyager* mission).

In this thesis, however, we are interested in studying CRs in the GC region. In such a faraway region, measuring the CR intensity directly is, of course, impossible. Therefore, we can only detect the products of their interactions with ambient matter and radiation. This can be done, most notably, through γ-ray observations, but as we will see in the following, other tracers can be used.

γ-ray observations may give us information about the current distribution of CRs in the region under examination, but not at which rate CRs are accelerated and *injected* into this region. Our ultimate interest lies in determining how much energy has to be injected in the form of CRs in order to explain a number of observations of the CMZ. To do so, we need to study the link between the *injection* and the *equilibrium* spectrum of CRs, which is constrained by observations. This link is provided by the CR transport equation.

The aim of this chapter is to describe the CR transport equation and to show how it can be solved by means of finite difference numerical methods.

3.1 The cosmic-ray transport equation

The CR transport equation was first derived by Parker (1965). A number of formal derivations have followed (Gleeson et al. 1967; Dolginov et al. 1968; Jokipii et al. 1967; Jokipii et al. 1970; Skilling 1975; Luhmann 1976; Berezinskii et al. 1990; Kirk 1994; Schlickeiser 2002; Zank 2014). A heuristic derivation of the CR transport equation was described by Moraal (2013), who showed that its physical meaning can be understood starting from the continuity equation

$$\frac{\partial n}{\partial t} = -\nabla \cdot \vec{j} + Q \tag{3.1}$$

that states that the number of CR particles is conserved. Here, n is the particle number density, $\vec{j}$ is the flux density of particles (the net rate of particles moving out of a unit

volume), and q is the generation (or destruction) of CR particles per unit volume per unit time. This equation states that the evolution in time of the number of particles in a unit of volume is equal to the rate of particles flowing in or out through the surface enclosing this volume plus the rate at which particles may be created or destroyed within this volume. Finally, as we are also interested in the distribution of particles in momentum (or energy), the equation has to be completed with a term expressing the divergence of the particle flux in momentum space (analogue to the term $\nabla \cdot \vec{j}$ describing spatial transport).

The general form of the transport equation for CR particles can be written as:

$$\frac{\partial f}{\partial t} = \vec{\nabla} \cdot \left(\bar{D} \vec{\nabla} f - \vec{v}_w f \right) + \frac{1}{p^2} \frac{\partial}{\partial p} \left(p^2 D_{\mathrm{pp}} \frac{\partial f}{\partial p} \right) - \frac{1}{p^2} \frac{\partial}{\partial p} \left[\dot{p} p^2 f - \vec{\nabla} \cdot \vec{v}_w \frac{p^3}{3} f \right] + Q(\vec{x}, p, t), \quad (3.2)$$

where $f(\vec{x}, p, t)$ is the CR particle distribution function at time t, position $\vec{x}$ and momentum p. $\bar{D}(\vec{x}, p)$ is the spatial diffusion tensor, $\vec{v}_w$ is the advection velocity, $D_{\mathrm{pp}}(\vec{x}, p)$ is the momentum diffusion coefficient and, $\dot{p}(\vec{x}, p)$ is the momentum loss rate. $Q(\vec{x}, p, t)$ is the source term.

In the remainder of this section, we shall describe very briefly the different physical processes that affect the propagation of CRs. The interested reader is referred to Schlickeiser (2002) and Strong et al. (2007) for more details and references on the topics.

The first term on the right-hand side of Eq. 3.2 describes the spatial transport of CRs. The two terms in the parenthesis refer to spatial diffusion in a turbulent interstellar magnetic field and to advection in a Galactic wind of velocity $\vec{v}_w$.

The second term refers to diffusion in momentum space. This is due to the fact that the magnetic perturbations that scatter particles are not at rest but move at a characteristic velocity, and therefore, not only the particle pitch angle but also its momentum is modified during scattering.

The third term describes systematic changes in the momentum of particles, either due to interactions with the ambient matter of radiation (first term in the parenthesis) or to adiabatic compression/decompression (second term).

Finally, the fourth term accounts for the injection of freshly accelerated particles in the region.

3.1.1 The transport equation in the Galactic Centre

We want to model the transport of CRs in the CMZ. Because of the approximate cylindrical shape of this region, it is convenient to adopt a cylindrical symmetry. Then, the relevant spatial coordinates are the radial distance r in the GP from the GC and the vertical distance or height z from the GP. Additionally, we assume that the advection velocity $\vec{v}_w$ is perpendicular to the GP. Under such assumptions, the CR transport equation in the CMZ becomes:

$$\begin{aligned} \frac{\partial f}{\partial t} &= D_{\mathrm{rr}} \frac{\partial^2 f}{\partial r^2} + \left(\frac{D_{\mathrm{rr}}}{r} + \frac{\partial D_{\mathrm{rr}}}{\partial r} \right) \frac{\partial f}{\partial r} + D_{\mathrm{zz}} \frac{\partial^2 f}{\partial z^2} + \frac{\partial D_{\mathrm{zz}}}{\partial z} \frac{\partial f}{\partial z} - v_{\mathrm{w}} \frac{\partial f}{\partial z} \\ &+ \frac{1}{p^2} \frac{\partial}{\partial p} \left(p^2 D_{\mathrm{pp}} \frac{\partial f}{\partial p} \right) + \frac{p}{3} \frac{\partial v_{\mathrm{w}}}{\partial z} \frac{\partial f}{\partial p} - \frac{1}{p^2} \frac{\partial}{\partial p} (\dot{p} p^2 f) + Q \end{aligned} \quad (3.3)$$

where D_{rr} and D_{zz} are the spatial diffusion coefficients along the radial and vertical axes, respectively.

The CR transport equation is a second-order parabolic partial differential equation (PDE). Due to the complexity of the equation, the preferred method of solving it is *numerically*. There are various approaches to solving the transport equation numerically. In the following, we explore a grid-based method similar to PICARD (Kissmann 2014), `DRAGON2` (Evoli et al. 2017) and GALPROP (Moskalenko et al. 1998; Strong et al. 1998).

3.2 Numerical resolution of partial differential equations

The CR transport equation in cylindrical coordinates (Eq. 3.3) gives the particle distribution function at a distance r from the GC, at a height z from the GP for a particle momentum p at time t. When the injection is time-dependent, the solution never sets into a stationary regime. In this case, we are obliged to solve the time integral of the equation. In the case where the injection is continuous in time, it is possible to solve for $\frac{\partial f}{\partial t} = 0$. However, computationally, the first method is preferred also when searching for steady-state solutions.

Following previous literature, and in particular Evoli et al. (2017), we numerically solve Eq. 3.3 using a Crank-Nicolson scheme (Crank et al. 1947). Instead of solving for the CR distribution function $f(r, z, p, t)$, we solve for the CR density per total momentum p

$$N(r, z, p, t) \equiv 4\pi p^2 f(r, z, p, t). \tag{3.4}$$

3.2.1 Discretisation and finite differences method

As the name suggests, the grid-based method involves a grid in the coordinates r, z, p and t. These points are marked r_i, z_j, p_k and t_n respectively. The CR density at a given grid point (i, j, k, n) is $N(r_i, z_j, p_k, t_n)$ which will be denoted as:

$$N^n_{i,j,k} \tag{3.5}$$

On such a grid, the derivative operators are replaced by finite differences. A finite difference can be a *forward*, *backward* or *central* approximation.

First derivative

If $u(x)$ is a function assumed to be smooth, the first derivative of this function evaluated at a certain point x_0 is $u'(x_0)$. Using a *forward* difference operator D_+, this value can be approximated as:

$$D_+ u(x)|_{x_0} = \frac{u(x_0 + h) - u(x_0)}{h} \tag{3.6}$$

Using a *backward* difference operator D_-, the value can be approximated as:

$$D_- u(x)|_{x_0} = \frac{u(x_0) - u(x_0 - h)}{h} \tag{3.7}$$

These expressions provide a first-order accurate (in h) approximation of the first derivative. In order to obtain a second-order approximation, a Taylor expansion of $u(x_0 + h)$ and $u(x_0 - h)$ can be used:

$$u(x_0 + h) \simeq u(x_0) + hu'(x_0) + \frac{h^2}{2}u''(x_0) + O(h^3) \tag{3.8}$$

$$u(x_0 - h) \simeq u(x_0) - hu'(x_0) + \frac{h^2}{2}u''(x_0) + O(h^3) \tag{3.9}$$

We obtain a second-order approximation of the first derivative by subtracting Eq.3.9 from Eq.3.8. Such an approximation corresponds to the *central* difference scheme:

$$u'(x_0) = Du(x)|_{x_0} = \frac{u(x_0 + h) - u(x_0 - h)}{2h} \tag{3.10}$$

The most appropriate choice between *forward*, *backward* and *central* difference schemes depends on the physical model under consideration. For the CR density N_m at a given point x_m on an irregularly spaced grid, the three difference schemes read:

$$D_+ N_m = \frac{N_{m+1} - N_m}{x_{m+1} - x_m} \tag{3.11}$$

$$D_- N_m = \frac{N_m - N_{m-1}}{x_m - x_{m-1}} \tag{3.12}$$

$$D N_m = \frac{N_{m+1} - N_{m-1}}{x_{m+1} - x_{m-1}}. \tag{3.13}$$

Second derivative

We adopt a *central* difference approximation for the second derivative. We can obtain the difference scheme by adding Eq.3.8 and Eq.3.9:

$$u''(x_0) = D^2u(x)|_{x_0} = \frac{u(x_0 + h) - 2u(x_0) + u(x_0 - h)}{h^2} \tag{3.14}$$

It is worth noting that $D^2u(x) = D_+D_-u(x) = D_-D_+u(x)$. As for the CR density N_m at a given point x_m, such a scheme reads:

$$D^2N_m = \frac{2}{x_{m+1} - x_{m-1}} \left(\frac{N_{m+1} - N_m}{x_{m+1} - x_m} - \frac{N_m - N_{m-1}}{x_m - x_{m-1}} \right) \tag{3.15}$$

3.2.2 Operator splitting

The transport equation is a PDE where each term represents a different physical process. Each physical process affects the spatial or momentum distribution of the particle, which in turn makes the local particle density vary over time. The *operator-splitting* method involves separating the PDE into several parts such that the solution at the end of a timestep depends on intermediate solutions obtained by solving the PDEs with each operator separately.

Consider a differential equation of the form:

$$\frac{du}{dt} = \mathcal{L}_1(u) + \mathcal{L}_2(u) \tag{3.16}$$

where $\mathcal{L}_1$ and $\mathcal{L}_2$ are linear functions of u such that $\mathcal{L}_1 = l_1 u$ and $\mathcal{L}_2 = l_2 u$.

The solution to this differential equation is:

$$u(t) = u(0)e^{(l_1+l_2)t} \tag{3.17}$$

Using the solution at $t = t_n$, it is possible to obtain the solution at $t = t_{n+1} = t_n + \Delta t$ as a function of the solution:

$$u(t_{n+1}) = u(t_n)e^{(l_1+l_2)\Delta t} \tag{3.18}$$

If the operators $\mathcal{L}_1$ and $\mathcal{L}_2$ commute, then this expression can be written as:

$$u(t_{n+1}) = u(t_n)e^{(l_1\Delta t)}e^{(l_2\Delta t)} \tag{3.19}$$

and even if they do not commute, this is still an approximate expression. The Baker-Campbell-Hausdorff formula states that the error is $\mathcal{O}(t^2)$ (Campbell 1897; Zhang 2017).

This is equivalent to first applying $\mathcal{L}_1$ to the solution at $t = t_n$ to get an intermediate solution u^* and then applying $\mathcal{L}_2$ to u^* or in the opposite order:

$$\begin{aligned} u^* &= u(t_n)e^{(l_1\Delta t)} \\ u(t_{n+1}) &= u^* e^{(l_2\Delta t)} \end{aligned} \tag{3.20}$$

In addition to operator splitting, we also implement *dimensional splitting* where we divide a single spatial operator further according to the number of spatial dimensions.

3.2.3 Crank-Nicolson scheme

In this section, we present a semi-implicit method called the Crank-Nicolson scheme (Crank et al. 1947; Press et al. 1992). This scheme is unconditionally stable even for large timesteps and is second-order accurate in time.

We only deal with one operator at a time using the operator-splitting method. Let us consider a sample PDE for CR density N that depends on a coordinate $x(x_i)$ and time $t(t_n)$, with a single operator $\tilde{\mathcal{O}}$ and injection Q divided by the number of total operators n_{os}:

$$\frac{\partial N}{\partial t} = \tilde{\mathcal{O}}(N) + \frac{Q}{n_{\text{os}}} \tag{3.21}$$

An *explicit* discrete form of this equation only has terms at time t_n on the right-hand side. It reads:

$$\frac{N_i^{n+1} - N_i^n}{\Delta t} = [\tilde{\mathcal{O}}(N)]_i^n + \frac{Q_i}{n_{\text{os}}} \tag{3.22}$$

where we assumed for simplicity that the injection term Q is stationary. On the contrary, an *implicit* discretisation only has terms at time t_{n+1} on the right-hand side:

$$\frac{N_i^{n+1} - N_i^n}{\Delta t} = [\tilde{\mathcal{O}}(N)]_i^{n+1} + \frac{Q_i}{n_{\text{os}}} \tag{3.23}$$

Both these schemes are only first-order accurate in time. We choose, instead, to use a Crank-Nicolson scheme, which is *semi-implicit*:

$$\frac{N_i^{n+1} - N_i^n}{\Delta t} = \frac{1}{2}\left([\tilde{\mathcal{O}}(N)]_i^n + [\tilde{\mathcal{O}}(N)]_i^{n+1}\right) + \frac{Q_i}{n_{\text{os}}} \tag{3.24}$$

Because both sides of the equality are centred on the timestep $n + \frac{1}{2}$, this method is second-order accurate in time.

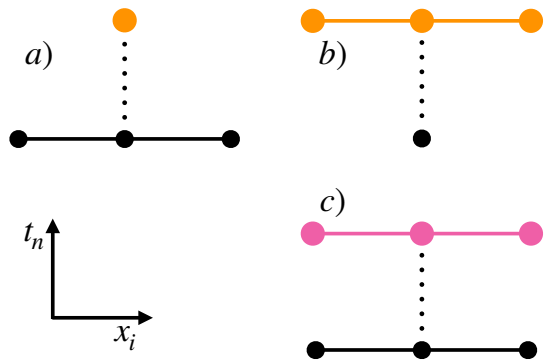

Figure 3.1: Schematic view of a fully explicit scheme (a), a fully implicit scheme (b) and a Crank-Nicolson scheme (c). The black and coloured points represent the known values at timestep t_n and the values at timestep t_{n+1}, respectively.

Following Evoli et al. (2017), we consider the most general form for the operator $\tilde{\mathcal{O}}$:

$$[\tilde{\mathcal{O}}(N)]_i^n = L_i N_{i-1}^n - C_i N_i^n + U_i N_{i+1}^n \tag{3.25}$$

Using this expression, Eq. 3.24 becomes:

$$-\frac{\Delta t}{2} L_i N_{i-1}^{n+1} + \left(1 + \frac{\Delta t}{2} C_i\right) N_i^{n+1} - \frac{\Delta t}{2} U_i N_{i+1}^{n+1} = \tag{3.26}$$

$$\frac{\Delta t}{2} L_i N_{i-1}^n + \left(1 - \frac{\Delta t}{2} C_i\right) N_i^n + \frac{\Delta t}{2} U_i N_{i+1}^n + \frac{\Delta t}{n_{\text{os}}} Q_i \tag{3.27}$$

This forms a tridiagonal set of linear equations that need to be solved for each operator consecutively at each timestep. The L_i, C_i and U_i coefficients, whose derivation is detailed

in Appendix A, are determined for each of the operators in the CR transport equation (Eq. 3.3) and reported in Table 3.1. In particular, we have an operator for radial diffusion, one for vertical diffusion and advection, and one for reacceleration.

The case of momentum losses (including adiabatic losses) has been treated in a different way, as illustrated in Appendix A. Indeed, for such an operator, following Evoli et al. (2017) and Kissmann (2014) we found a discretisation scheme by integrating the equation

$$\frac{\partial N}{\partial t} = \frac{\partial}{\partial p}(\dot{p}N) + \frac{Q}{n_{\rm os}} \tag{3.28}$$

between momentum p_k and p_{k+1}. We still get a tridiagonal set of linear equations but the definition of the L, C and U coefficients reported in Table 3.1 is different compared to that of the Crank-Nicolson scheme of Eq. 3.26:

$$(1 - \Delta t U_k) N_{k+1}^{n+1} + (1 + \Delta t C_k) N_k^{n+1} = \tag{3.29}$$

$$(1 + \Delta t U_k) N_{k+1}^{n} + (1 - \Delta t C_k) N_k^{n} + \frac{\Delta t}{n_{\rm os}}(Q_{k+1} + Q_k) \tag{3.30}$$

Although the Crank-Nicolson method is stable for any choice of the time step Δt, this quantity would still determine the accuracy of the solution. The choice of Δt should, therefore, be small enough to record the time evolution accurately. Moreover, in order to be sure that the steady-state solution is reached, the total runtime $t_{\rm tot}$ needs to be long enough for the system to reach equilibrium.

Here, we are interested in a vast range of particle energies (keV-PeV for electrons and MeV-PeV for protons), and the corresponding characteristic timescales of the problem vary by many orders of magnitude in such broad energy domains. This implies that if we set Δt small enough to reproduce the shortest characteristic time scale of the problem, the number of timesteps needed to reach steady state ($t_{\rm tot}/\Delta t$) will be exceedingly large, leading to impractically long computation times.

In order to mitigate this problem, we adopt an approach also used in `DRAGON2` code (see Evoli et al. 2017, and references therein), based on a progressive decrease of the timestep from a maximum value $\Delta t_{\rm max}$ to a minimum value $\Delta t_{\rm min}$. Each timestep between these two limits is used m times. Once a timestep Δt has been used m times, the following timestep will be $\Delta t/1.1$. In the following, we explain how $\Delta t_{\rm min}$ and $\Delta t_{\rm max}$ are chosen.

For each propagation process contained in the transport equation, we determine the time required for particles to move between two adjacent grid points in space or momentum. This duration depends on the considered process, grid spacing and particle momentum. We compute the minimum timestep for each process, and the smallest of these values must be the minimum timestep for the entire scheme. This global minimum timestep is constrained by the loss times in low momenta and the diffusion times in very high momenta.

The total run time $t_{\rm tot}$ is estimated by focussing on the typical timescales for each process. Instead of looking at the time taken to move across one grid space, we compute the time necessary for particles to be removed from the system. For example, the diffusion timescale is the time necessary for a particle to diffuse out of the CMZ. We choose $t_{\rm tot}$ such that all processes have had the time to make a significant impact on the particles. The relation

Crank-Nicolson scheme for the CR transport equation

Process	**L**	**C**	**U**	**B.C**
r-diffusion	$\frac{2D_i}{\Delta r_c \Delta r_d} - \frac{D_i}{r_i \Delta r_c} - \frac{D_{i+1}-D_{i-1}}{\Delta r_c^2}$	$\frac{2D_i}{\Delta r_c}\left(\frac{1}{\Delta r_u} + \frac{1}{\Delta r_d}\right)$	$\frac{2D_i}{\Delta r_c \Delta r_u} + \frac{D_i}{r_i \Delta r_c} + \frac{D_{i+1}-D_{i-1}}{\Delta r_c^2}$	$N_{-1} = N_0$
z-diffusion	$\frac{2D_j}{\Delta z_c \Delta z_d} - \frac{D_{j+1}-D_{j-1}}{\Delta z_c^2}$	$\frac{2D_j}{\Delta z_c}\left(\frac{1}{\Delta z_u} + \frac{1}{\Delta z_d}\right)$	$\frac{2D_j}{\Delta z_c \Delta z_u} + \frac{D_{j+1}-D_{j-1}}{\Delta z_c^2}$	$N_{-L/L} = 0$
z-advection $z < 0$	0	$-\frac{v_j}{\Delta z_u}$	$-\frac{v_{j+1}}{\Delta z_u}$	$N_{-L/L} = 0$
z-advection $z = 0$	$\frac{v_{j-1}}{\Delta z_c}$	0	$-\frac{v_{j+1}}{\Delta z_c}$	$N_{-L/L} = 0$
z-advection $z > 0$	$\frac{v_{j-1}}{\Delta z_d}$	$\frac{v_j}{\Delta z_d}$	0	$N_{-L/L} = 0$
losses	0	$\frac{\dot{p}_k}{\Delta p_u}$	$\frac{\dot{p}_{k+1}}{\Delta p_u}$	$N_{p_{\max}} = 0$
reacceleration	$-\frac{D_{k+1}-D_{k-1}}{\Delta p_c^2} + \frac{2D_k}{\Delta p_c \Delta p_d} + \frac{2D_{k-1}}{p_{k-1}\Delta p_d}$	$\frac{2D_k}{\Delta p_c}\left(\frac{1}{\Delta p_u} + \frac{1}{\Delta p_d}\right) + \frac{2D_k}{p_k \Delta p_d}$	$\frac{D_{k+1}-D_{k-1}}{\Delta p_c^2} + \frac{2D_k}{\Delta p_c \Delta p_u}$	$N_0 = \frac{p_0^2}{p_1^2} N_1$ and $N_{p_{\max}} = 0$

Table 3.1: Numerical scheme where $\Delta s_u = s_{i+1} - s_i$, $\Delta s_d = s_i - s_{i-1}$ and $\Delta s_c = s_{i+1} - s_{i-1}$

between the different timesteps and the total run time is:

$$\sum_{\Delta t=\Delta t_{\min}}^{\Delta t=\Delta t_{\max}} m\Delta t \geq t_{\text{tot}} \tag{3.31}$$

$$\frac{\Delta t_{\max}}{\Delta t_{\min}} \approx 10^{4-5} \tag{3.32}$$

$\Delta t_{\max}$ and m are chosen such that the total time is at least a few factors higher than the estimated time needed to reach equilibrium. However, in order to avoid problems of convergence, $\Delta t_{\max}$ should be at most 4-5 orders of magnitude higher than $\Delta t_{\min}$. The value of $\Delta t_{\max}$ should be chosen with some care. Otherwise, the guess solution obtained with large Δt could be too different from the steady-state solution, and one would need too many timesteps for the system to reach convergence, thus nullifying the method's benefits in terms of computation time.

As mentioned above, the timesteps are constrained by the solution at very low and very high energies. Hence, $\Delta t_{\min}$ may be longer than the global minimum timescale for all processes as long as the numerical solution has an acceptable level of accuracy. This also applied for the total run time t_{tot}, and therefore to $\Delta t_{\max}$. They may be shorter or longer than the estimation from timescales for different processes as long as the numerical solution at a later time does not differ too much.

It is very simple to test whether the chosen timesteps are sufficiently short or long. In the very low energies, we expect the losses to dominate over other processes. Hence, the solution becomes analytical in the low-energy domain. This is also the case in the high-energy domain as diffusion becomes dominant. Once we have chosen all the physical parameters, we verify that for a given $\Delta t_{\min}$, $\Delta t_{\max}$ and m, the solution at the two ends of the energy domain match the analytical solution from losses and diffusion. This will count as a validation of the choice of timesteps.

The rationale behind such a method is the following. Consider the total time, t_{tot}, required for the system to reach the steady state. We evolve the system beyond t_{tot} using $\Delta t \gg \Delta t_{\min}$ for most of the time. This is possible thanks to the unconditional stability of the Crank-Nicolson scheme. This phase of the calculation consists of approximating the final steady-state solution. The subsequent temporal evolution, performed with a progressively decreasing timestep and finally, the steps with $\Delta t = \Delta t_{\min}$, are needed to bring the system to converge with good accuracy to the "true" steady-state solution. Such an approach combines a substantial reduction of the number of timesteps, compared to simply taking $\Delta t = t_{\text{tot}}/\Delta t_{\min}$, with a good accuracy of the result.

Notice also that this method can only be used when a steady-state solution exists for the problem at hand. In the case of unsteady solutions, as in the case of an impulsive source, we rely on the parallelisation of the loops over spatial and momentum coordinates to accelerate the code.

3.3 Validation of the CR transport scheme

In the context of numerical resolution of PDEs, stability and convergence are necessary qualities of an algorithm. Due to truncation errors, the solution at different timesteps may

differ from the "true" solution. Throughout the iteration, the algorithm can amplify these errors by accumulating them. In the case of a *stable* algorithm, however, these errors are effectively damped so that the numerical solution converges to the "true" solution. Moreover, a slight variation in the initial conditions for a stable algorithm does not make the numerical solution diverge from the "true" solution.

A numerical scheme is said to be *consistent with order p* when the local truncation error τ approaches 0 as Δt^p approaches 0. The Lax-Richtmyer stability theorem requires that the norm of the matrix used in the iteration remains less than or equal to 1. When a well-posed initial-value problem is consistent, the Lax-Richtmyer equivalence theorem states that the Lax-Richtmyer stability of an algorithm is necessary and sufficient for its convergence (Lax et al. 1956).

The CR transport equation is a well-posed initial value problem. The Crank-Nicolson scheme has been used in many disciplines and has been thoroughly studied. In particular, it has been shown that the Crank-Nicolson scheme is consistent with order 2 and that it is Lax-Richtmyer stable (Qu et al. 2017; Oishi et al. 2014; Fadugba et al. 2012). We can, therefore, apply the Lax-Richtmyer equivalence theorem to conclude that the Crank-Nicolson scheme is convergent.

The Crank-Nicolson scheme is stable and convergent, but it is not certain that it will converge to an *accurate* solution. To verify the accuracy of the numerical scheme, we compare the numerical solution to the analytical solution for simplified transport equations, where we only consider one process at a time. We use the test cases used in Evoli et al. (2017). We define $\tilde{\epsilon}$ the relative error between the analytical solution $N_{\rm an}$ and the numerical solution $N_{\rm num}$:

$$\tilde{\epsilon} = \frac{|N_{\rm an} - N_{\rm num}|}{N_{\rm an}} \tag{3.33}$$

Our scheme is considered accurate when $\tilde{\epsilon}$ is lower than 10 %. We consider different values for the total physical time $t_{\rm tot}$ to ensure our code converges to a steady-state solution by taking different values of m. We also vary the number of points in the grid and the minimum timestep $\Delta t_{\rm min}$ for the time evolution. The maximum timestep is then $10^5 \Delta t_{\rm min}$.

The choices for these parameters depend on the physical process. In this section, we test out the code by considering several problems that have analytical solutions. We choose the parameters to prove the convergence of the scheme to an accurate solution.

3.3.1 Test case for spatial diffusion

The two-dimensional diffusion equation is:

$$-D_{\rm zz}\frac{\partial^2 N}{\partial z^2} - \left[\frac{D_{\rm rr}}{r} + \frac{\partial D_{\rm rr}}{\partial r}\right]\frac{\partial N}{\partial r} - D_{\rm rr}\frac{\partial^2 N}{\partial r^2} = Q(r,z) \tag{3.34}$$

where r goes from 0 to R and z from -L to L. The boundary conditions are $N(r = R, z = \pm L) = 0$ and $\partial N/\partial r(r = 0) = 0$.

The injection function is taken to be:

$$Q(r,z) = \left(\frac{D_{\rm rr}}{R^2} + \frac{D_{\rm zz}}{L^2}\right)\frac{\pi^2 N_a}{4} + \frac{\pi D_0}{2R^2}\left(\frac{R}{r} + \frac{3r}{R}\right)\sin\left(\frac{\pi r}{2R}\right)\cos\left(\frac{\pi z}{2L}\right) \tag{3.35}$$

and the diffusion coefficients are defined as:

$$D_{\rm rr}(r) = D_0 \left(1 + \frac{r^2}{R^2}\right) \qquad D_{\rm zz} = f D_0 \tag{3.36}$$

The analytical solution for Eq.(3.34) is then:

$$N_a(r, z) = \cos\left(\frac{\pi r}{2R}\right) \cos\left(\frac{\pi z}{2L}\right) \tag{3.37}$$

In the following, we take $D_0 = 10^{28}\mathrm{cm}^2\ \mathrm{s}^{-1}$ and $f = 0.1$. We examine the solution along r at $z = 0$ and $z = 200$ pc. In the same way, we examine the solution along z at $r = 1$ pc and $r = 200$ pc.

We first compare the solution at different total times for $\Delta t = 10^7$ s and a grid size of 10 points in r and 11 in z (Fig. 3.2). We find a total time of 10^{15} s is needed for the required accuracy ($\tilde{\epsilon} \leq 10^{-1}$). Next, we compare the solution obtained using different minimum timesteps $\Delta t_{\rm min}$ for a total time $t_{\rm tot} = 10^{15}$ s in a 10×11 grid in (r, z) (Fig. 3.3). We find that there is no significant difference between the three timesteps. Therefore, we choose the largest one $\Delta t_{\rm min} = 10^9$ s. Finally, we verify if the spatial resolution is sufficient. For a total time $t_{\rm tot} = 10^{15}$ s, and $\Delta t_{\rm min} = 10^9$s, the solutions are obtained in three different grids (Fig. 3.4). The number of points in r (z) are 10 (11), 20 (21) and 40 (41). We choose the grid size 40×41.

In order to check the number of points for z, we also look at the solution along z, using the same total time and $\Delta t_{\rm min}$ (Fig. 3.5). A 40×41 grid gives accurate results.

3.3.2 Test case for vertical diffusion and advection

An advection problem is unstable by definition. It is handled by adding numerical diffusion. Here, the operator corresponding to the z variable has both advection and vertical diffusion, so we treat both processes together:

$$-D_{\rm zz}\frac{\partial^2 N}{\partial z^2} + \frac{\partial v N}{\partial z} = Q_0 \delta(z) \tag{3.38}$$

The velocity is a discontinuous field where the norm is positive above the disk, negative below the disk, and null at $z = 0$, and the boundary condition is $N(z = \pm L) = 0$.

The solution to this equation is:

$$N(z) = N_0 \frac{1 - e^{w(1-|z|/L)}}{1 - e^w} \tag{3.39}$$

where w is found by solving the equation for $z > 0$:

$$w = -\frac{v_{\rm w} L}{D} \tag{3.40}$$

By integrating the equation in the neighbourhood of the Galactic disk (between 0^- and 0^+), the normalisation is:

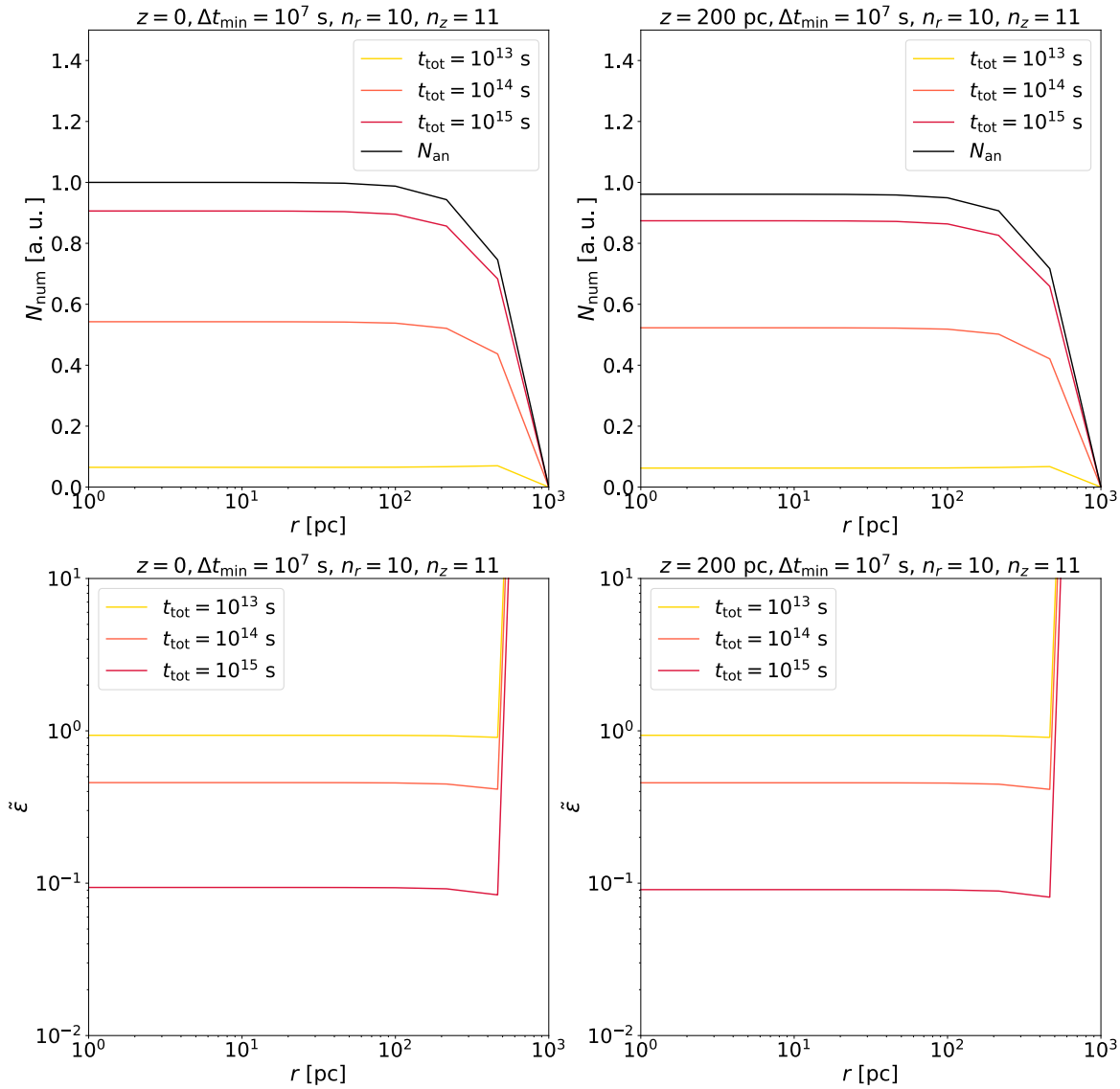

Figure 3.2: **Top:** Numerical solution compared to the analytical solution and **Bottom:** relative error as a function of radial distance r in the case of spatial diffusion at $z = 0$ and $z = 200$ pc at different times for $\Delta t_{\rm min} = 10^7$ s and grid size 10×11 in r and z.

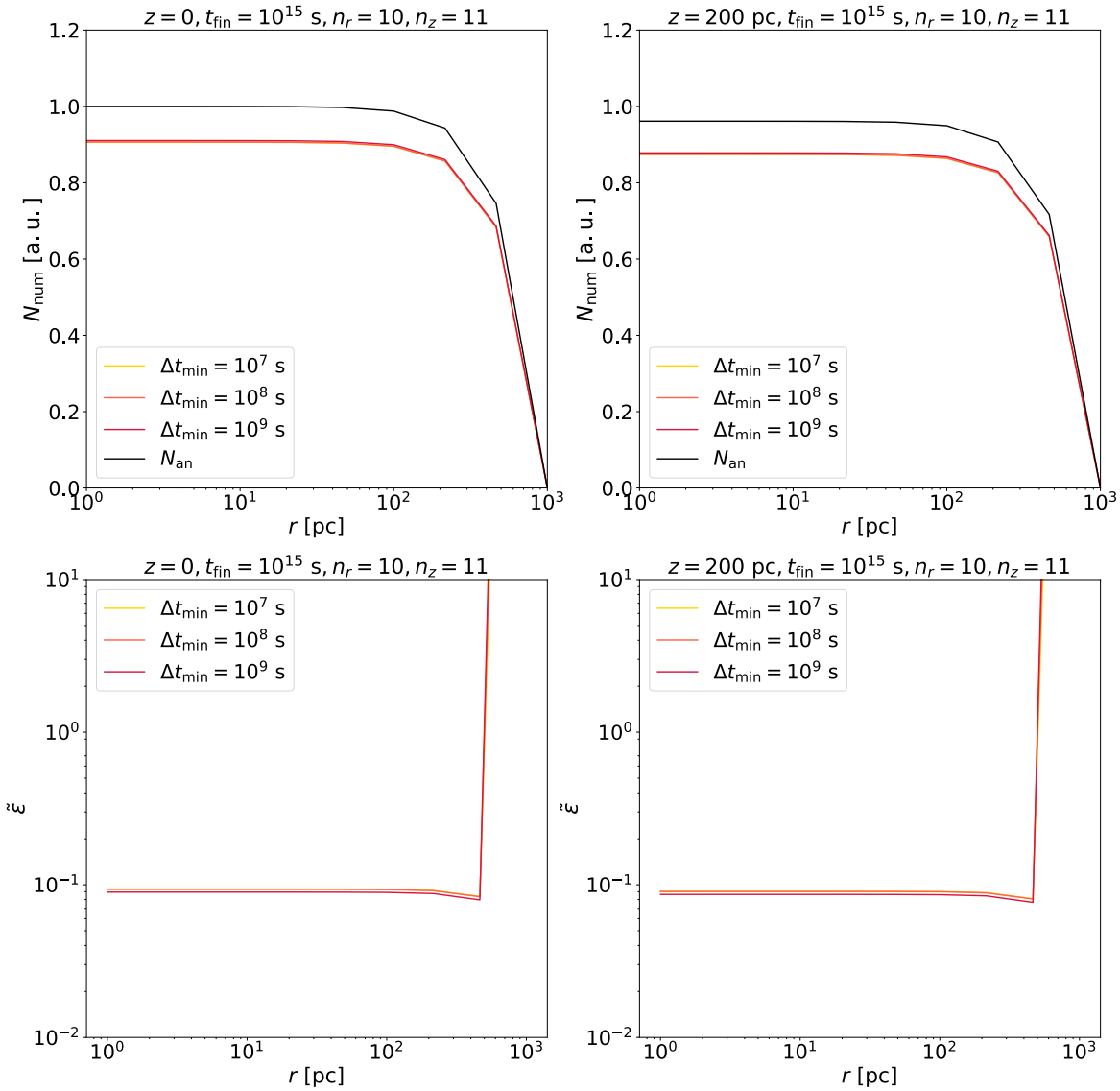

Figure 3.3: **Top:** Numerical solution compared to the analytical solution and **Bottom:** relative error as a function of radial distance r in the case of spatial diffusion at $z = 0$ and $z = 200$ pc for different $\Delta t_{\rm min}$ at total time $t_{\rm tot} = 10^{15}$ s and grid size 10×11 in r and z.

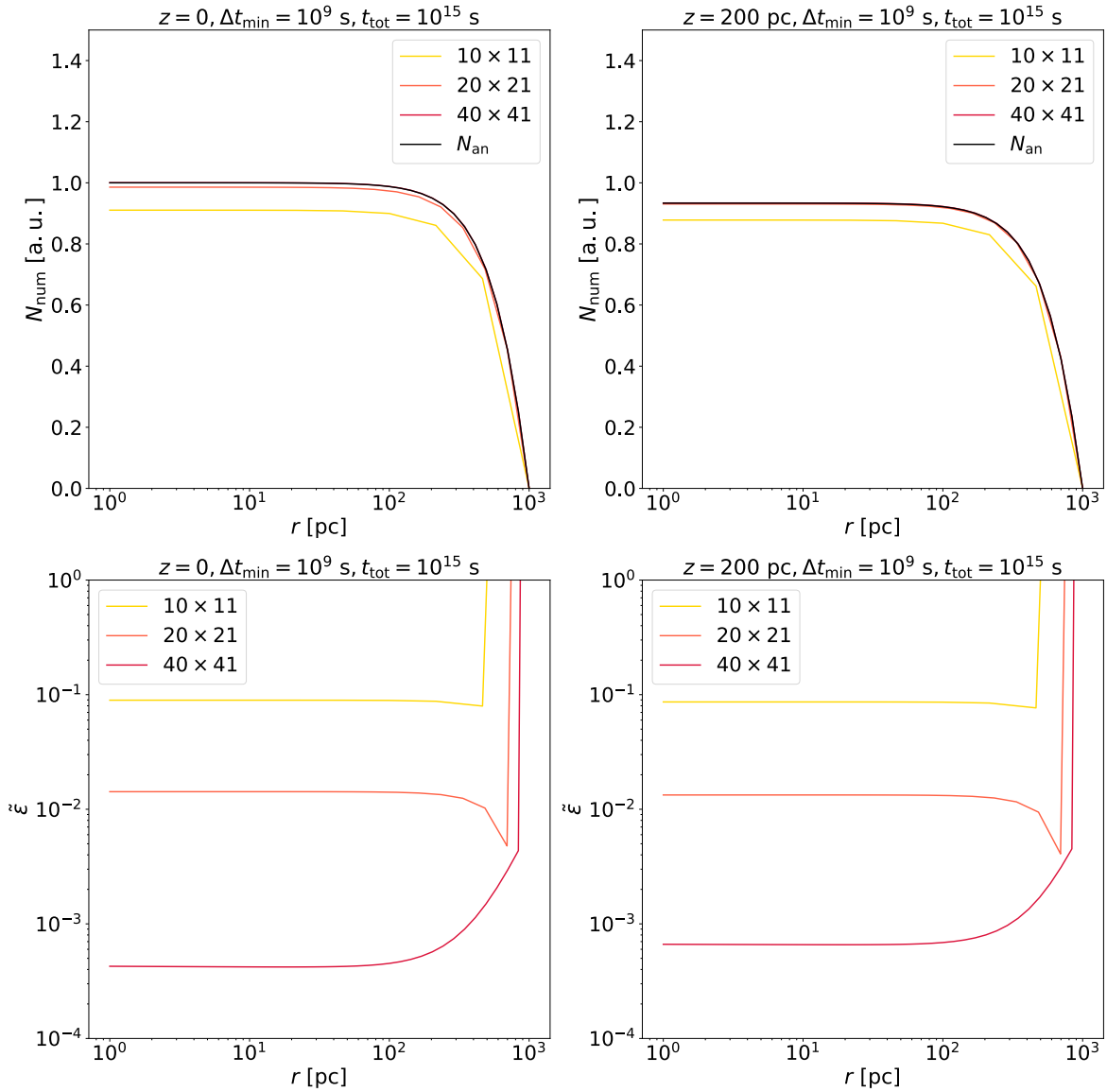

Figure 3.4: **Top:** Numerical solution compared to the analytical solution and **Bottom:** relative error as a function of radial distance r in the case of spatial diffusion at $z = 0$ and $z = 200$ pc for different grid sizes for $\Delta t_{\min} = 10^9$ s and total time $t_{\text{tot}} = 10^{15}$ s.

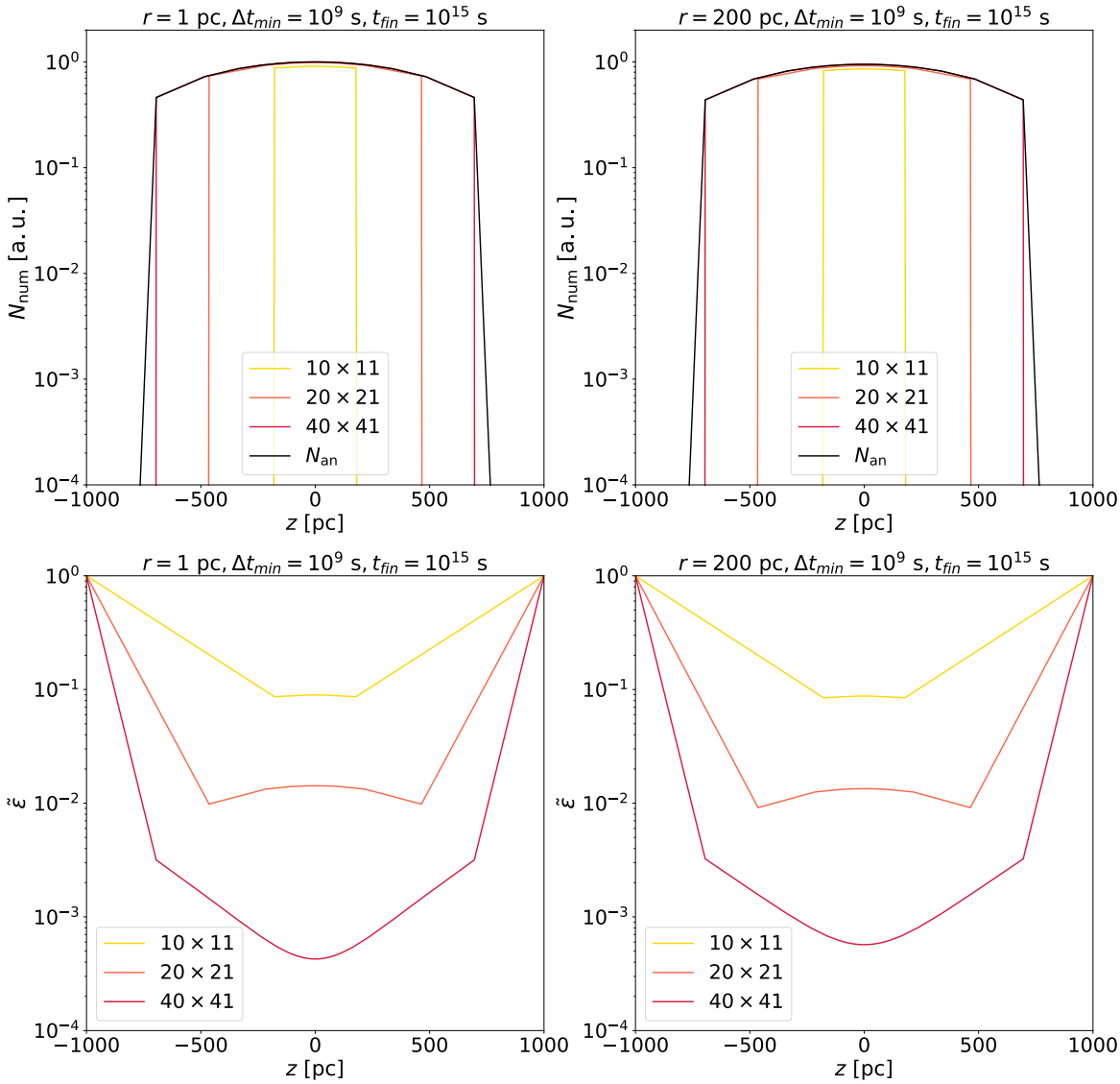

Figure 3.5: **Top:** Numerical solution compared to the analytical solution and **Bottom:** relative error as a function of vertical distance z in the case of spatial diffusion at $r = 1$ pc and $z = 200$ pc for different grid sizes for $\Delta t_{\mathrm{min}} = 10^9$ s and total time $t_{\mathrm{tot}} = 10^{15}$ s.

$$N_0 = \frac{Q_0}{2v_{\rm w}}(1 - e^w) \tag{3.41}$$

In a 10×11 grid and for $\Delta t_{\rm min} = 10^7$ s, we compare the solution at different total times $t_{\rm tot}$ (Fig. 3.6). Similarly to the diffusion case, we need a total time of 10^{15} s. Next, we compare the different minimum timesteps needed by using a total time of 10^{15} s and the same grid as above (Fig. 3.7). We choose $\Delta t_{\rm min} = 10^9$ s for the same reasons as for the diffusion case. Finally, we compare different grid sizes using $\Delta t_{\rm min} = 10^9$s and total time of 10^{15}s (Fig. 3.8). Again, similarly to the diffusion case, we choose 40×41.

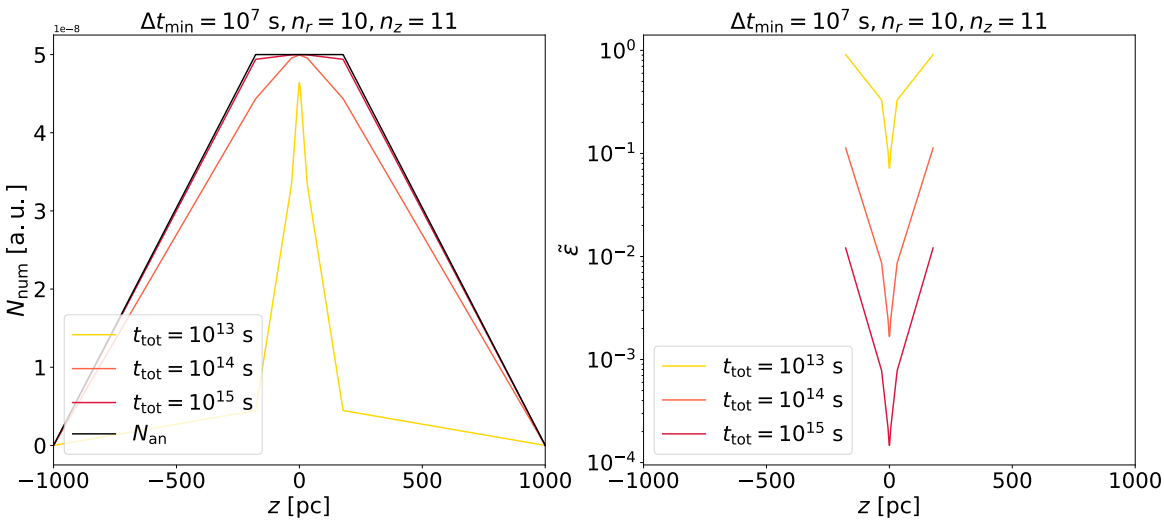

Figure 3.6: **Left:** Numerical solution compared to the analytical solution and **Right:** relative error as a function of vertical distance z in the case of vertical diffusion and advection at different total times for $\Delta t_{\rm min} = 10^7$ s and grid size 10×11 in r and z.

3.3.3 Test case for momentum losses

We take the loss equation:

$$-\frac{\partial}{\partial p}(\dot{p}N) = Q(p) \tag{3.42}$$

where $\dot{p} > 0$.

Integrating both sides between p and p_{max}, with the boundary condition $N(p_{max}) = 0$, the solution to this equation is:

$$N(p) = \frac{1}{\dot{p}} \int_p^{p_{max}} Q(p')dp' \tag{3.43}$$

Assuming a power-law for the injection spectrum of the form $Q(p) = Q_0 \left(\frac{p}{p_0}\right)^{-s}$, the solution becomes:

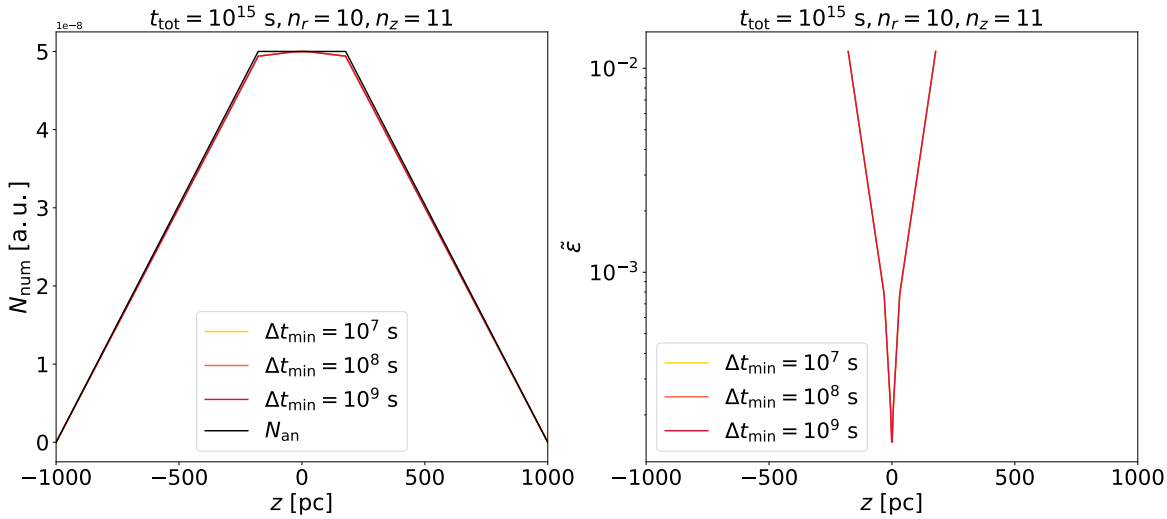

Figure 3.7: **Left:** Numerical solution compared to the analytical solution and **Right:** relative error as a function of vertical distance z in the case of vertical diffusion and advection for different minimum timesteps for a grid size 10×11 and a total time 10^{15} s.

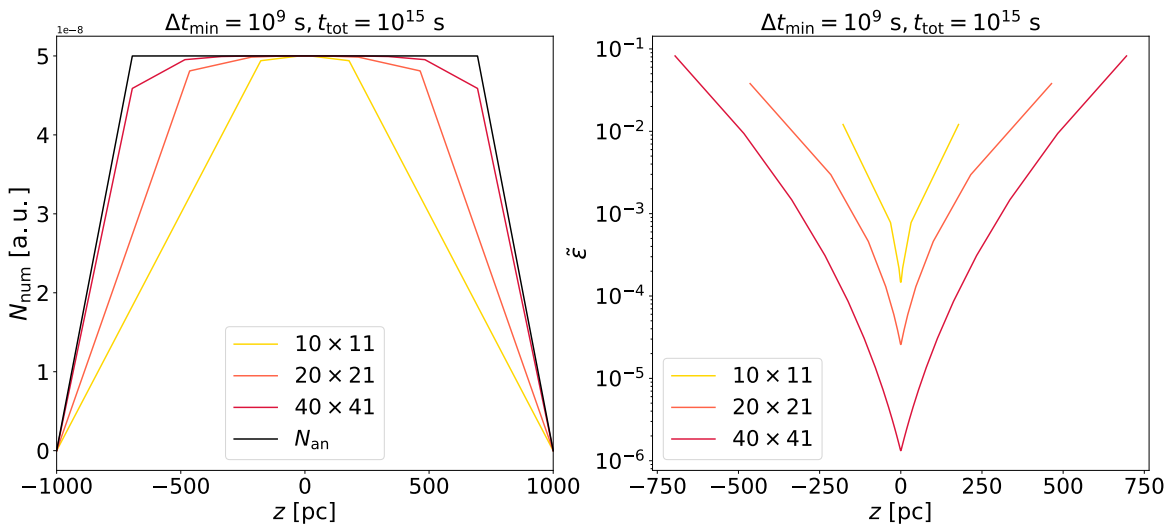

Figure 3.8: **Left:** Numerical solution compared to the analytical solution and **Right:** relative error as a function of vertical distance z in the case of vertical diffusion and advection for different grid sizes for $\Delta t = 10^9$ s and a total time 10^{15} s.

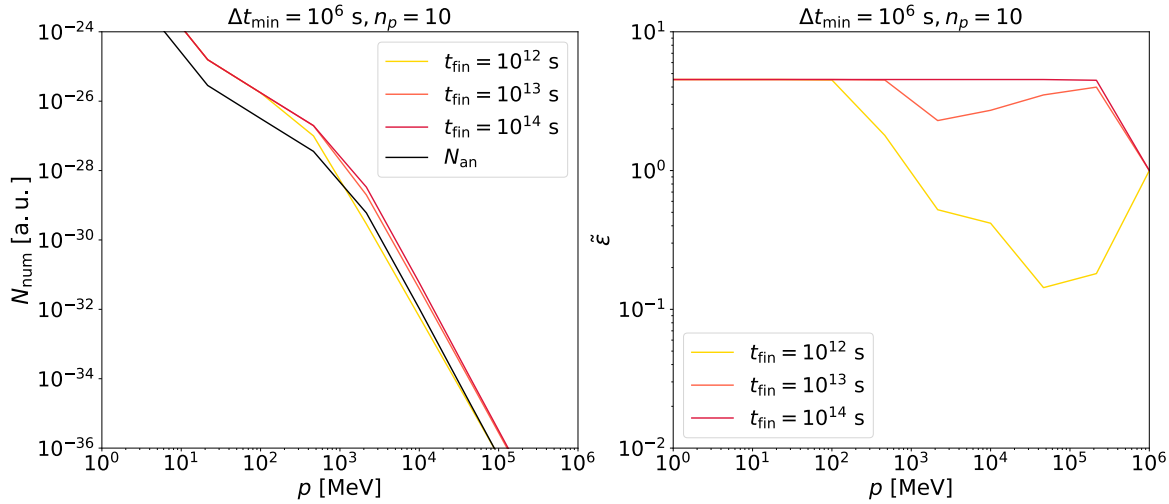

Figure 3.9: **Left:** Numerical solution compared to the analytical solution and **Right:** relative error as a function of momentum p in the case of momentum losses for different total times for $\Delta t = 10^6$ s and 40 points in p.

$$N(p) = \frac{Q_0 {p_0}^s}{\dot{p}} \left(\frac{p^{1-s}}{s-1} \right) \tag{3.44}$$

Following the same order as before, we compare the solution at different total times for a short minimum timescale $\Delta t_{\min} = 10^6$ s and 10 points in p (Fig. 3.9). We notice that the solution converges but not towards the analytical solution.

We chose $n_p = 40$ as the number of points because it is close to the number of points that worked for the spatial coordinates, which are $n_r = 40$ and $n_z = 41$. We take a longer total time to be certain that the steady state has been reached. We compare the numerical solutions using different minimum timesteps (Fig. 3.10). As there are no significant differences, we choose the longest timestep $\Delta t_{\min} = 10^9$ s.

However, the relative error is still above the acceptable limit. In order to rule out needing longer total times, we compare the numerical solution at different total times for $\Delta t_{\min} = 10^9$ s and $n_p = 40$ (Fig. 3.11). Longer total times do not improve the result.

Shorter minimum timesteps and longer total times do not improve the level of accuracy of the numerical solutions. Therefore, we increase the number of points in momentum p in an attempt to obtain a better solution. For $\Delta t_{\min} = 10^9$ s and $t_{\text{tot}} = 10^{15}$ s, we compare the solutions obtained in grids with different numbers of points in p (Fig. 3.12). With 80 points in p, the relative error is below 10%.

3.3.4 Test case for reacceleration

The equation for reacceleration is:

$$-\frac{\partial}{\partial p} \left[p^2 D_{\text{pp}} \frac{\partial}{\partial p} \left(\frac{N}{p^2} \right) \right] = Q(p) \tag{3.45}$$

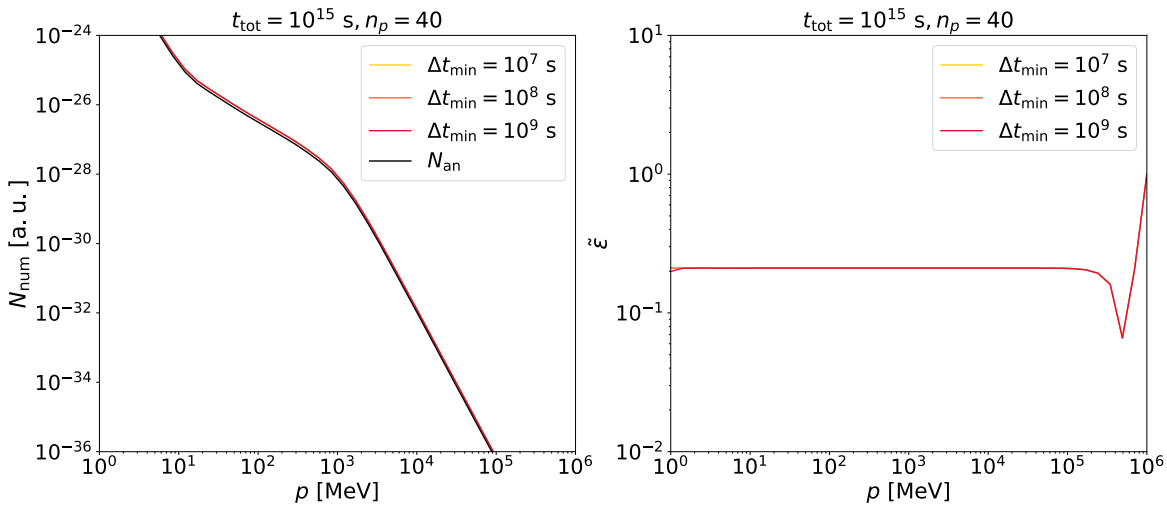

Figure 3.10: **Left:** Numerical solution compared to the analytical solution and **Right:** relative error as a function of momentum p in the case of momentum losses for different minimum timesteps for $t_{\text{tot}} = 10^{15}$ s and 10 points in p.

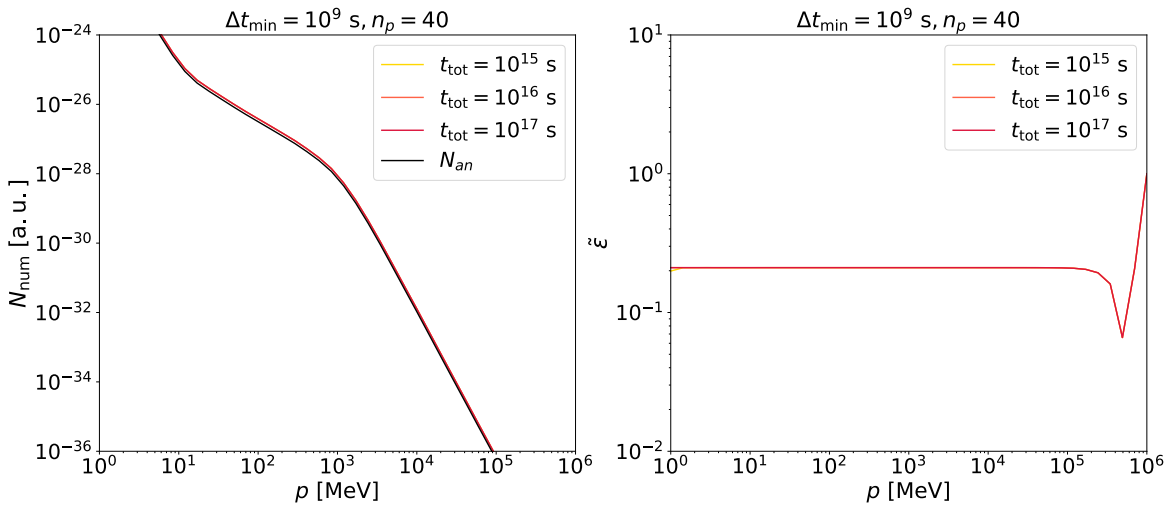

Figure 3.11: **Left:** Numerical solution compared to the analytical solution and **Right:** relative error as a function of momentum p in the case of momentum losses for different total times for $\Delta t = 10^9$ s and 40 points in p.

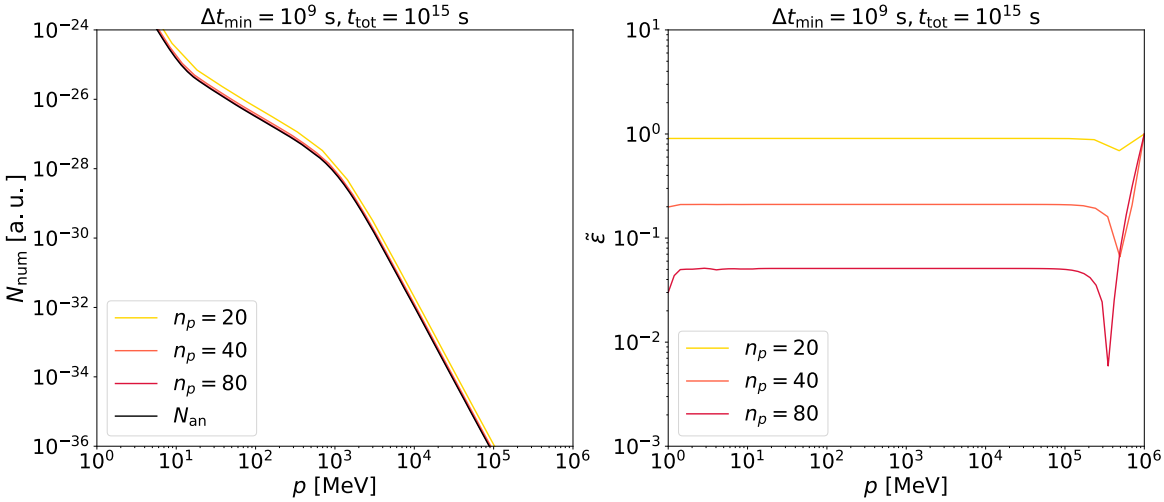

Figure 3.12: **Left:** Numerical solution compared to the analytical solution and **Right:** relative error as a function of momentum p in the case of momentum losses for different numbers of points in p for $\Delta t = 10^9$ s and $t_{\text{tot}} = 10^{15}$ s.

where the momentum diffusion coefficient is $D_{\text{pp}} = D_0 \left(\frac{p}{p_0}\right)^2$ and the injection is the same power-law with $s > 2$. The analytical solution in this case is:

$$N(p) = \frac{Q_0 p^2}{D_0(1-s)} \left[\frac{1}{s+2} \left(\left(\frac{p}{p_0}\right)^{-s-2} - \left(\frac{p_{\max}}{p_0}\right)^{-s-2} \right) + \frac{1}{3} \left(\frac{p_{\min}}{p_0}\right)^{-s+1} \left(\left(\frac{p_{\max}}{p_0}\right)^{-3} - \left(\frac{p}{p_0}\right)^{-3} \right) \right] \tag{3.46}$$

We compare the solutions for different total times for $\Delta t_{\min} = 10^{23}$ s and 10 points in p (Fig. 3.13). A total time of $t_{\text{tot}} = 5 \times 10^{28}$ s is sufficient for the numerical solution to converge. The solution is, however, very different from the analytical solution.

We verify the choice of the minimum timestep by comparing the numerical solution for different timesteps at a total time $t_{\text{tot}} = 5 \times 10^{28}$ s and 10 points in p (Fig. 3.14). We find that the numerical solution does not vary for the different timesteps. Therefore, we may continue with the longest timestep $\Delta t_{\min} = 10^{23}$ s.

Since the relative error between the numerical and analytical solution is still larger than the acceptable threshold, we improve the resolution of the grid. We compare the numerical solutions in grids with different numbers of points in p for $\Delta t_{\min} = 10^{23}$ s and $t_{\text{tot}} = 5 \times 10^{28}$ s (Fig. 3.15). We find that with 1000 points, the numerical solution may be accepted as accurate.

3.4 Conclusion

In this chapter, we have described the transport equation solver. We have tested the solver for each physical process. For well-chosen parameters, we are able to find a solution that is

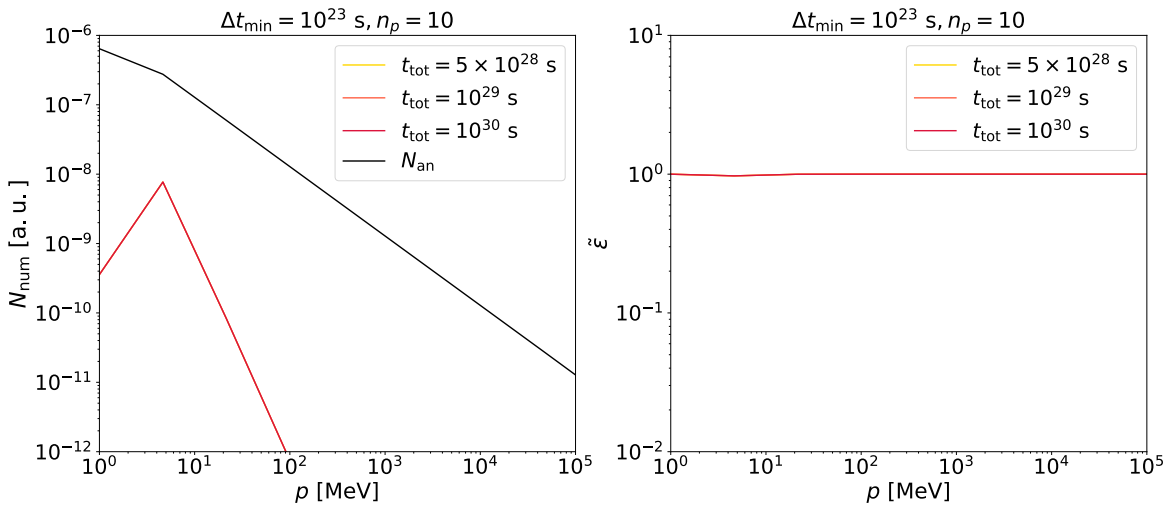

Figure 3.13: **Left:** Numerical solution compared to the analytical solution and **Right:** relative error as a function of momentum p in the case of reacceleration for different total times for $\Delta t_{\min} = 10^{23}$ s and 10 points in p.

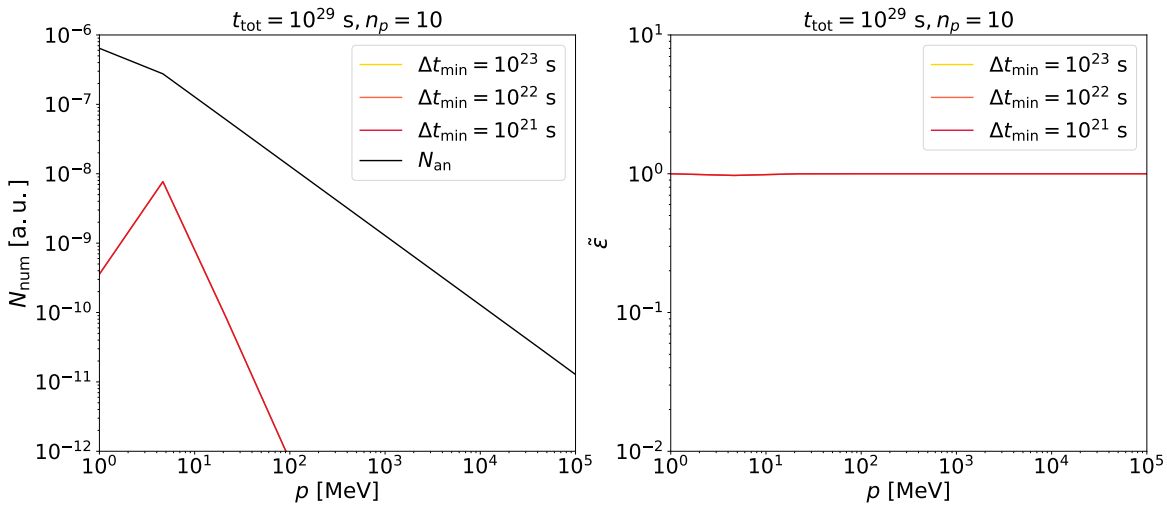

Figure 3.14: **Left:** Numerical solution compared to the analytical solution and **Right:** relative error as a function of momentum p in the case of momentum losses for different minimum timesteps for $t_{\text{tot}} = 5 \times 10^{28}$ s and 10 points in p.

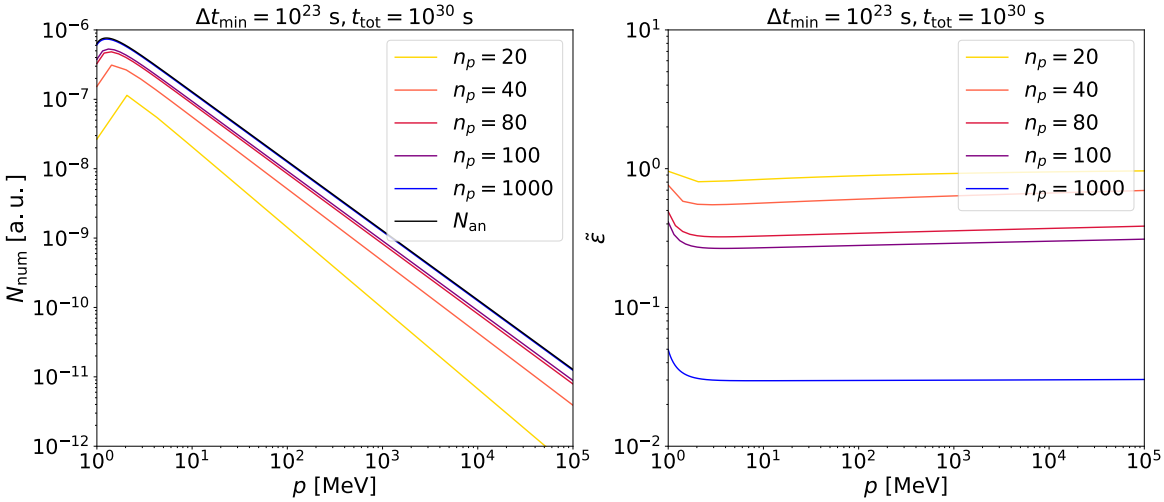

Figure 3.15: **Left:** Numerical solution compared to the analytical solution and **Right:** relative error as a function of momentum p in the case of momentum losses for different numbers of points in p for $\Delta t_{\rm min} = 10^{23}$ s and $t_{\rm tot} = 5 \times 10^{28}$ s.

acceptable (less than 10% error). In the following, we use this code to derive the solution to the CR transport equation.